

Challenge (Habanero)

- Q1.** A hiker's fuel consumption is modeled by 2^x , where x is the number of hours hiked. If she hiked for 3 hours, her fuel consumption would be
(A) $2^3 = 8$
(B) $2^2 = 4$
(C) $2^4 = 16$
(D) $2^5 = 32$
(E) $2^6 = 64$
- Q2.** Simplify $(3^2)^4$ to get
(A) 3^4
(B) 3^6
(C) 3^8
(D) $3^{2 \cdot 4} = 3^8$
(E) $3^{2+4} = 3^6$
- Q3.** A population of rabbits grows according to 4^x , where x is the number of years. If after 2 years, the population is 4^2 , what is the population after 5 years?
(A) 4^3
(B) 4^5
(C) 4^7
(D) $4^{2+3} = 4^5$
(E) $4^{5-2} = 4^3$
- Q4.** Simplify $\frac{2^4}{2^2}$ to get
(A) 2^2
(B) 2^4
(C) 2^6
(D) $2^{4-2} = 2^2$
(E) $2^{4+2} = 2^6$
- Q5.** If $x^3 = 64$, what is x ?
(A) $x = 2$
(B) $x = 3$
(C) $x = 4$
(D) $x = 5$
(E) $x = 6$
- Q6.** The volume of a hiking backpack is 2^x cubic feet, where x is the number of compartments. If the backpack has 4 compartments, what is its volume?
(A) $2^2 = 4$
(B) $2^3 = 8$
(C) $2^4 = 16$
(D) $2^5 = 32$
(E) $2^6 = 64$
- Q7.** Simplify $3^2 \cdot 3^4$ to get
(A) 3^4
(B) 3^6
(C) 3^8
(D) $3^{2+4} = 3^6$
(E) $3^{2 \cdot 4} = 3^8$
- Q8.** A mountain's altitude increases by 2^x feet, where x is the number of miles hiked. If after 3 miles, the altitude is $2^3 = 8$ feet, what is the altitude after 6 miles?
(A) $2^4 = 16$
(B) $2^5 = 32$
(C) $2^6 = 64$
(D) $2^{3+3} = 2^6 = 64$
(E) $2^{6-3} = 2^3 = 8$

Open Ended Questions (FRQ)

- Q9.** A group of friends want to go on a hiking trip. The cost of the trip is 2^{x+2} dollars per person, where x is the number of people going. If 4 friends are going, what is the total cost of the trip?
- Q10.** Simplify the expression $\frac{5^2 \cdot 5^3}{5^4}$ and explain the steps you took to simplify it.

Chef's Tips

- Tip 1: Make sure students understand the rules of exponents before attempting the test
- Tip 2: Encourage students to use the rules of exponents to simplify expressions before solving
- Tip 3: Remind students that the test is timed, so they should manage their time effectively